EXP NO: 7 EXPERIMENT TO UNDERSTAND EDA-QUANTITATIVE AND QUALITATIVE ANALYSIS

AIM:

Understand the importance of EDA-Quantitative and Qualitative analysis.

PYTHON CODE:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

%matplotlib inline

np.random.seed(42)

n = 250

shopping = pd.DataFrame({
   "Purchase_Amount": np.random.randint(200, 5000, n),
   "Discount": np.random.randint(0, 50, n),
   "Age": np.random.randint(18, 60, n),
   "Gender": np.random.choice(["Male", "Female"], n),
   "Category": np.random.choice(
      ["Electronics", "Clothing", "Home", "Sports"], n),
   "Payment": np.random.choice(
      ["UPI", "Card", "Cash"], n),
   "Delivery": np.random.choice(
      ["Standard", "Express"], n)
})

shopping["Final_Price"] = shopping["Purchase_Amount"] * (1 - shopping["Discount"]/100)

print(shopping.head())
print(shopping.info())
print(shopping.describe())
sns.jointplot(
   x="Discount",
   y="Final_Price",
   data=shopping
)
```

```
plt.show()
sns.jointplot(
    x="Age",
    y="Purchase_Amount",
    data=shopping,
    kind="hex"
)
plt.show()
plt.figure(figsize=(6,4))

sns.heatmap(
    shopping.corr(numeric_only=True),
    annot=True,
    cmap="coolwarm"
)

plt.title("Correlation Heatmap")
plt.show()
```

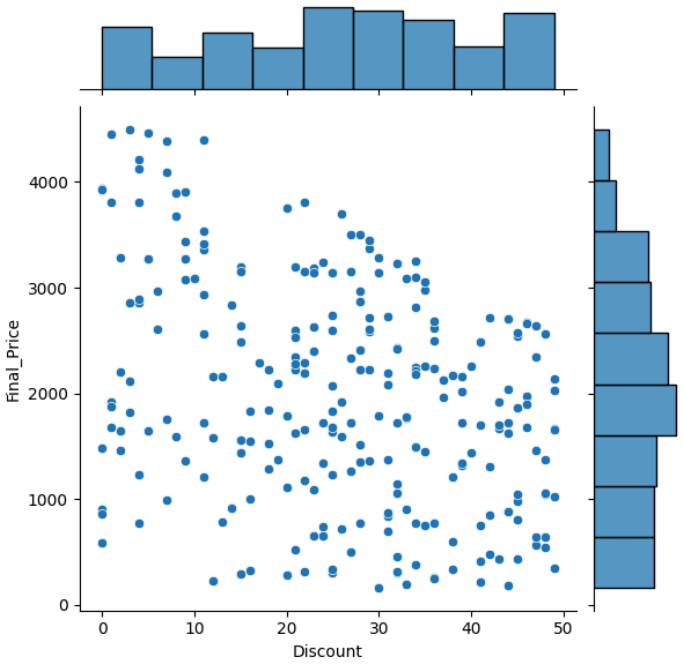
OUTPUT:

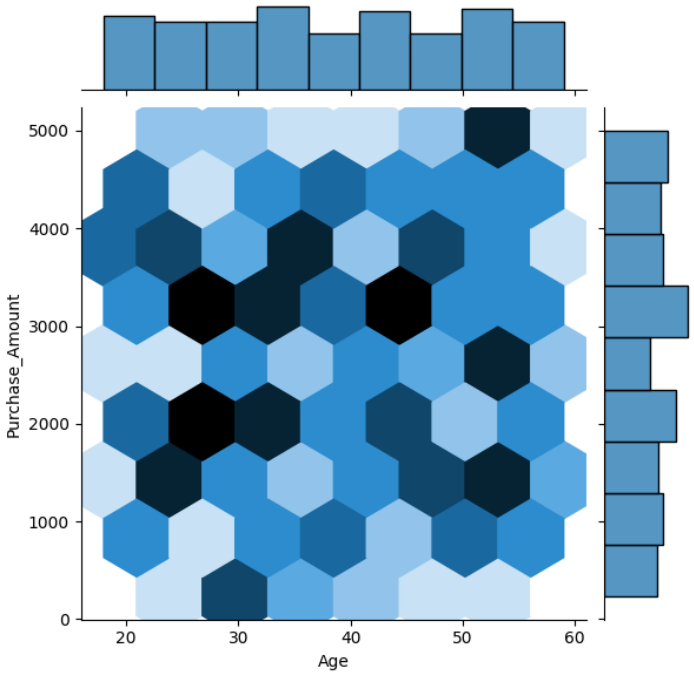
	Purchase_Amount	Discount	Age	Gender	Category	Payment	Delivery	\
0	1060	47	53	Male	Clothing	Cash	Express	
1	3972	11	35	Female	Sports	Card	Express	
2	3292	21	56	Male	Electronics	UPI	Express	
3	666	21	49	Female	Home	Cash	Standard	
4	4626	45	41	Male	Electronics	Card	Standard	

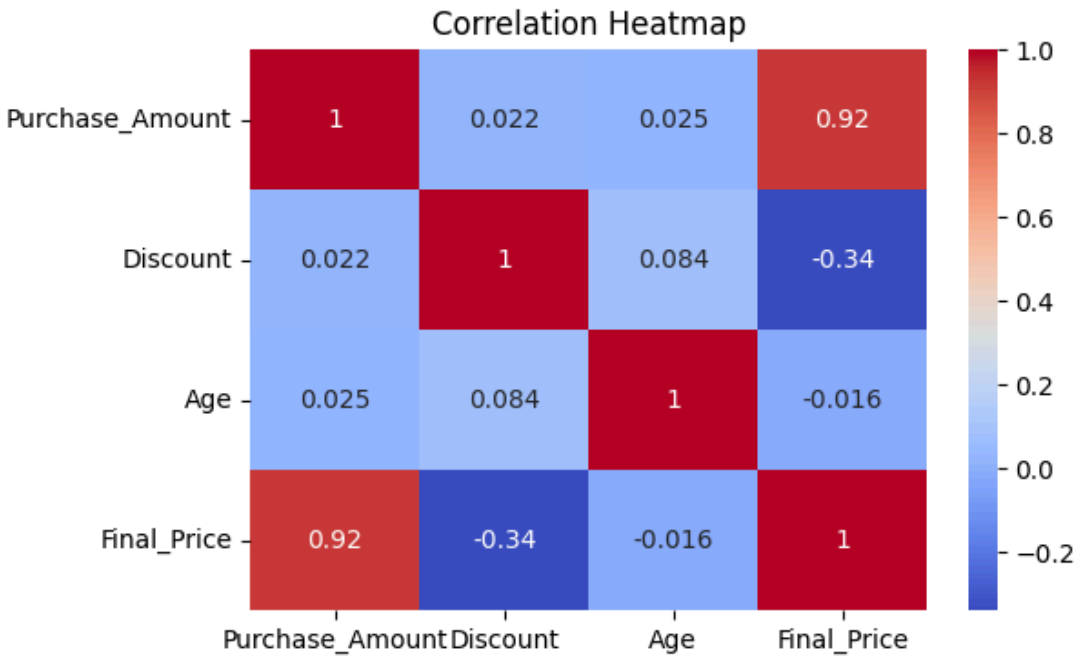
	Final_Price
0	561.80
1	3535.08
2	2600.68
3	526.14
4	2544.30

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 250 entries, 0 to 249
Data columns (total 8 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   Purchase_Amount  250 non-null    int64
 1   Discount         250 non-null    int64
 2   Age              250 non-null    int64
 3   Gender           250 non-null    object
 4   Category         250 non-null    object
 5   Payment          250 non-null    object
 6   Delivery         250 non-null    object
 7   Final_Price      250 non-null    float64
dtypes: float64(1), int64(3), object(4)
memory usage: 15.8+ KB
None
```

	Purchase_Amount	Discount	Age	Final_Price
count	250.00000	250.000000	250.000000	250.000000
mean	2669.57200	25.964000	38.524000	1972.322680
std	1348.01858	14.057352	11.860358	1061.107602
min	234.00000	0.000000	18.000000	163.800000
25%	1564.50000	15.000000	29.000000	1152.510000
50%	2763.00000	27.000000	38.000000	1887.300000
75%	3759.75000	36.000000	49.000000	2701.040000
max	4998.00000	49.000000	59.000000	4494.010000







RESULT:

Thus the code is executed successfully.